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List of paper publications

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1	New stability indicating RP-HPLC method development and validation for simultaneous estimation of Emtricitabine, Tenofovir, Disoproxil Fumarate and efavirenz with degradation kinetics.	K.S.Sumanth, A.Srinivasarao, D.Gowrishankar	International Journal of all research Education and scientific methods (IJARESM)	3-4



New Stability Indicating RP-HPLC Method Development and Validation for Simultaneous Estimation of Emtricitabine, Tenofovir Disoproxil Fumarate and Efavirenz with Degradation Kinetics

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ABSTRACT

A new, simple, rapid and stability-indicating RP-HPLC method was developed for the estimation of Emtricitabine (EMT), Tenofovir disoproxil fumarate (TDF) and Efavirenz (EFA) in pharmaceutical dosage forms and validated. The HPLC method was developed on SHISEIDO C18 column (250 x 4.6 mm i.d, 5 μ) using Acetonitrile: 20 MM phosphate buffer (pH 5.0 $\pm$ 0.1) in the ratio of 60:40 v/v at 256 nm. The retention times for EMT, TDF and EFA were found to be 2.4, 3.2 and 9.4 min respectively. Linearity was established in the range of 4 – 20 μ g/ml, 6-30 μ g/ml and

12 – 60 μ g/ml for EMT, TDF and EFA respectively. The method was precise with %RSD < 2 for both intraday and interday precision. The accuracy of the method was performed over three levels of concentration and the recovery was in the range of 98-102%. The drugs were individually subjected to forced degradation (thermal, photolytic, hydrolytic, and oxidative stress conditions) studies for seven days. EMT and EFA showed maximum degradation in acidic medium, TDF showed maximum degradation in basic medium. The method was successfully applied for quantifying the drugs in marketed dosage forms.

Keywords: Emtricitabine, Forced degradation, Phosphate buffer, RP-HPLC, Tenofovir disoproxil fumarate.



INTRODUCTION

Emtricitabine (EMT) is a nucleoside reverse transcriptase inhibitor (NRTI). Chemically it is 5-fluoro-1-(2R, 5S)-[2-(hydroxymethyl)-1, 3-oxathiolan-5-yl] cytosine (Fig. 1). EMT is the (-) enantiomer of thio analog of cytidine which differs from other cytidine analogs, in that it has fluorine in the 5th position. EMT is an antiviral agent used for the prevention of perinatal HIV-1 reverse transcriptase [1]. It is also active against Hepatitis B virus [2]. Tenofovir disoproxil fumarate (TDF) is a fumaric acid salt of the bis- isopropoxy carbonyl oxymethyl ester derivative of tenofovir. TDF is a nucleotide reverse transcriptase inhibitor. TDF is converted to Tenofovir, an acyclic nucleoside Phosphonate (Nucleotide) analog of adenosine 5'-Monophosphate. Chemically it is 9-[(R)-2-[bis[(isopropoxycarbonyl) oxy] methoxy] phosphinyl] methoxy] propyl] adenine fumarate [3] (Physician's Desk Reference, 2008) (Fig. 2).

Efavirenz (6-chloro- 4 - (2 - Cyclopropyl - 1 - ethynyl) - 4 - trifluoro methyl (4S) - 1, 4 - dihydro - 2H - benzo [D] [1,3] oxazin - 2-one) is a non-nucleoside reverse transcriptase inhibitor (NNRTI) and is used as part of highly active antiretroviral therapy (HAART) for the treatment of a human immunodeficiency virus (HIV) type 1[1,2]. For HIV infection that has not previously been treated, Efavirenz and Lamivudine in combination with Zidovudine or Tenofovir is the preferred NNRTI-based regimen. Efavirenz is also used in combination with other antiretroviral agents as part of an expanded post exposure prophylaxis regimen to prevent HIV transmission for those exposed to materials associated with a high risk for HIV transmission [4,5] (Fig. 3)

The present study is to develop a stability indicating RP-HPLC method for EMT, TDF and EFA. The objective of the study is to subject the drugs for acid, base, peroxide, light, thermal degradation and estimate its extent of degradation in different days. A Literature survey reveals that several analytical methods for the estimation of EMT, TDF and EFA based on UV